

GLCM

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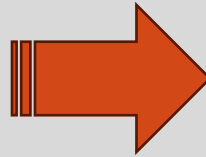
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OUTLINE

LAST WEEK (WEEK#12):

- Distance Transform
- Watershed Segmentation
- Evaluation in Medical Image Analysis



THIS WEEK (WEEK#13):

- Texture Based Feature Extraction : GLCM



TEXTURE-BASED FEATURE EXTRACTION

GLCM



FEATURE IN IMAGE

1. Color
2. Edges
3. Corners
4. Texture

APPLICATION AREA OF TEXTURE ANALYSIS

- Food processing industry
- Medical image analysis
- Biometrics analysis (fingerprints, iris or retina, etc.)
- Global information system (land analysis, etc.)

METHODS FOR TEXTURE FEATURE ANALYSIS

1. Statistical
 - a) General statistical parameters
 - b) Autocorrelation features
 - c) Laws texture energy features
 - d) **Co-occurrence matrix-based features**
 - e) Local Binary Pattern (LBP) features
2. Structural
3. Model
 - a) Fractal features
 - b) Random fields features
4. Filter
 - a) Gabor filters
 - b) Wavelet

CO-OCCURRENCE MATRIX-BASED FEATURES

- It is a matrix of frequencies at which $\rightarrow$ two pixels, separated by a certain vector, occur in the image

- Co-occurrence matrix is defined as,

$$p_{ij}(\Delta_x, \Delta_y) = WQ(i, j | \Delta_x, \Delta_y)$$

- Where,

$$W = \frac{1}{(M - \Delta_x)(N - \Delta_y)}$$
$$Q(i, j | \Delta_x, \Delta_y) = \sum_{n=1}^{N-\Delta_y} \sum_{m=1}^{M-\Delta_x} A$$

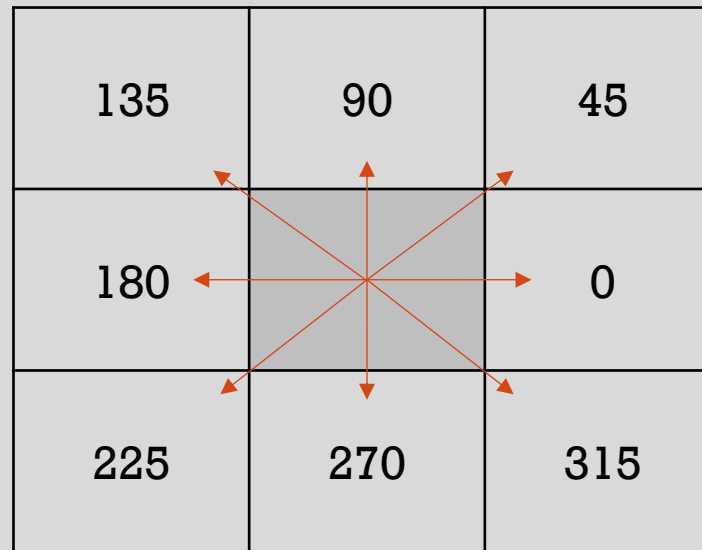
- Where

$$A = \begin{cases} 1 & \text{if } f(m, n) = i \text{ and } f(m + \Delta_x, n + \Delta_y) = j \\ 0 & \text{otherwise} \end{cases}$$

COMPUTATION OF CO-OCCURRENCE MATRIX

- It has $N \times N$ (N = Number of gray-values) i.e., the rows & columns represent the set of possible pixel values $\rightarrow$ Gral level Co-occurrence Matrix (GLCM)
- It is computed based on two parameters:
- $d \rightarrow$ Relative distance between the pixel pair (measured in pixel number e.g., 1,2, ...)
- $\theta \rightarrow$ Relative orientation / rotational angle. (e.g., 0° , 45° , 90° , 135° , ...)

8 DIRECTIONS/ORIENTATIONS (θ) OF ADJECENCY



- We consider θ as horizontal (0°), front diagonal (45°), vertical (90°), and back diagonal (135°)

COMPUTATION OF CO-OCCURRENCE MATRIX

Image matrix

0	0	1	1
0	0	1	1
0	2	2	2
2	2	3	3

Pixel values: 0,1,2,3. So, $N = 4$

So, size of CM = 4x4

$$d = 1$$

$\theta = \text{horizontal } (0^\circ)$

Find the **number of co-occurrences** of pixel i to the neighboring pixel value j

i,j	0	1	2	3
0	#(0,0)	#(0,1)	#(0,2)	#(0,3)
1	#(1,0)	#(1,1)	#(1,2)	#(1,3)
2	#(2,0)	#(2,1)	#(2,2)	#(2,3)
3	#(3,0)	#(3,1)	#(3,2)	#(3,3)

EXAMPLE: COMPUTATION

$d = 1$

$\theta = \text{horizontal } (0^\circ)$

Image

0	0	1	1
0	0	1	1
0	2	2	2
2	2	3	3



2	2	1	0

CM for Image

i,j	0	1	2	3
0	#(0,0)	#(0,1)	#(0,2)	#(0,3)
1	#(1,0)	#(1,1)	#(1,2)	#(1,3)
2	#(2,0)	#(2,1)	#(2,2)	#(2,3)
3	#(3,0)	#(3,1)	#(3,2)	#(3,3)

EXAMPLE: COMPUTATION

$d = 1$

$\theta = \text{horizontal } (0^\circ)$

Image

0	0	1	1
0	0	1	1
0	2	2	2
2	2	3	3



2	2	1	0
0	2	0	0
0	0	3	1
0	0	0	1

CM for Image

i,j	0	1	2	3
0	#(0,0)	#(0,1)	#(0,2)	#(0,3)
1	#(1,0)	#(1,1)	#(1,2)	#(1,3)
2	#(2,0)	#(2,1)	#(2,2)	#(2,3)
3	#(3,0)	#(3,1)	#(3,2)	#(3,3)

EXAMPLE: COMPUTATION

$d = 1$

$\theta = \textit{vertical} (90^\circ)$

Image

0	0	1	1
0	0	1	1
0	2	2	2
2	2	3	3



CM for Image

i,j	0	1	2	3
0	#(0,0)	#(0,1)	#(0,2)	#(0,3)
1	#(1,0)	#(1,1)	#(1,2)	#(1,3)
2	#(2,0)	#(2,1)	#(2,2)	#(2,3)
3	#(3,0)	#(3,1)	#(3,2)	#(3,3)

EXAMPLE: COMPUTATION

$d = 1$

$\theta = \textit{vertical} (90^\circ)$

Image

0	0	1	1
0	0	1	1
0	2	2	2
2	2	3	3



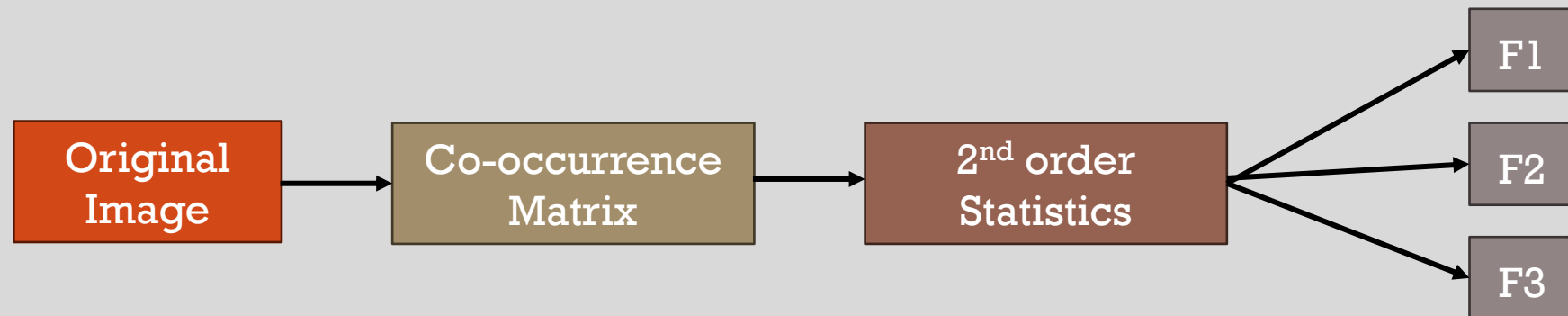
3	0	2	0
0	2	2	0
0	0	1	2
0	0	0	0

CM for Image

i,j	0	1	2	3
0	#(0,0)	#(0,1)	#(0,2)	#(0,3)
1	#(1,0)	#(1,1)	#(1,2)	#(1,3)
2	#(2,0)	#(2,1)	#(2,2)	#(2,3)
3	#(3,0)	#(3,1)	#(3,2)	#(3,3)

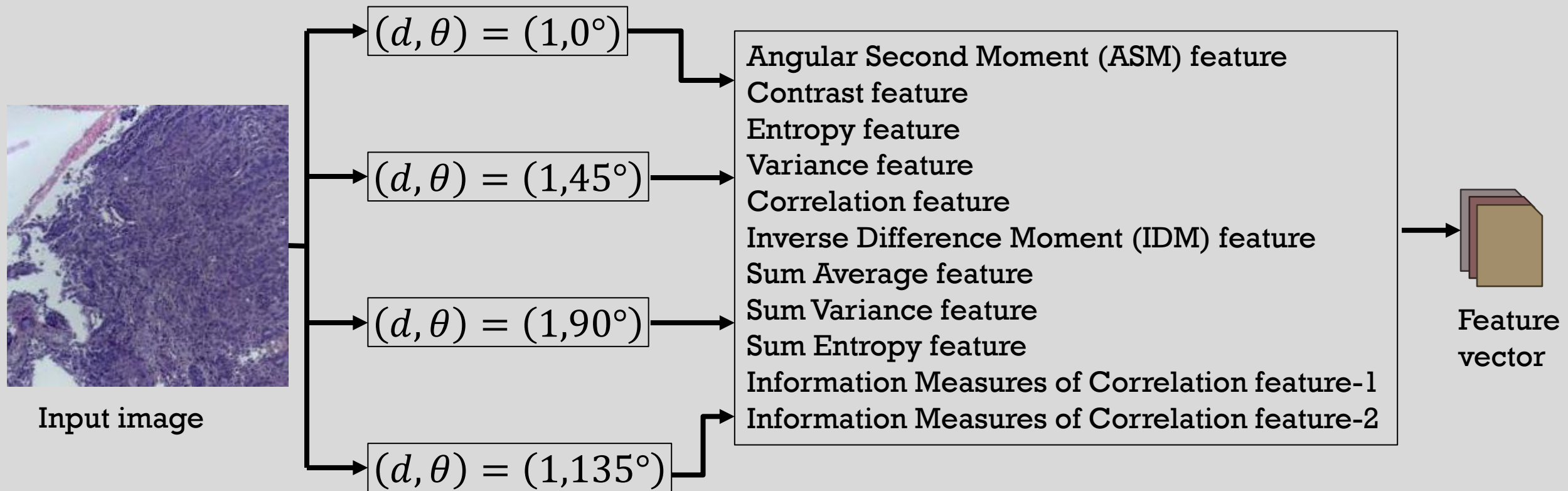
FEATURES ON CO-OCCURRENCE MATRIX

- Co-occurrence matrices capture properties of a texture
- But they are not directly useful for further analysis (e.g., comparison of two textures)



- **11 Numeric features** are computed from a matrix

FEATURES ON CO-OCCURRENCE MATRIX



FEATURES ON CO-OCCURRENCE MATRIX

Angular Second Moment (ASM) → Energy

- Definisi:
 - Ukuran keseragaman (uniformity) dalam citra. ASM tinggi berarti distribusi tekstur seragam atau berpola kuat

$$ASM = \sum_{i,j} P(i,j)^2$$

- Interpretasi:
 - Nilai tinggi → Citra halus/ teratur
 - Nilai rendah → Citra kasar atau bervariasi

Contrast

- Definisi:
 - Mengukur perbedaan local didalam citra: apakah piksel-piksel dalam lingkungan memiliki perbedaan intensitas yang besar

$$Contrast = \sum_{i,j} (i,j)^2 \cdot P(i,j)$$

- Interpretasi:
 - Nilai tinggi → Tekstur kasar (banyak perbedaan tajam)
 - Nilai rendah → Tekstur halus

FEATURES ON CO-OCCURRENCE MATRIX

Entropy

- Definisi:
 - Mengukur kerandoman/ kompleksitas informasi pada tekstur

$$Entropy = - \sum_{i,j} P(i,j) \cdot \log_2(P(i,j) + \epsilon)$$

- Interpretasi:
 - Nilai tinggi → kompleks dan acak
 - Nilai rendah → pola teratur

Variance

- Definisi:
 - Mengukur disperse (penyebaran) nilai intensitas dari rata-rata

$$Variance = \sum_{i,j} (i - \mu)^2 \cdot P(i,j)$$

- Interpretasi:
 - Nilai tinggi → banyak variasi tekstur
 - Nilai rendah → homogen

FEATURES ON CO-OCCURRENCE MATRIX

Correlation

- Definisi:

- Mengukur hubungan linier antara pasangan piksel: apakah nilai satu piksel dapat diprediksi dari tetangganya

$$\text{Correlation} = \frac{\sum_{i,j} (i - \mu_i)(j - \mu_j)P(i,j)}{\sigma_i \sigma_j}$$

- Interpretasi:

- Nilai mendekati 1 → korelasi kuat
- Nilai mendekati 0 → tidak berkorelasi

Inverse Difference Moment (IDM) → Homogeneity

- Definisi:

- Mengukur kesamaan nilai piksel: semakin mirip nilai piksel tetanggan, semakin tinggi IDM

$$IDM = \sum_{i,j} \frac{P(i,j)}{1 + (i - j)^2}$$

- Interpretasi:

- Nilai tinggi → tekstur halus dan mirip
- Nilai rendah → tekstur kasar

FEATURES ON CO-OCCURRENCE MATRIX

Sum Average

- Definisi:
 - Rata-rata dari jumlah indeks baris dan kolom (i+j) pada GLCM

$$SumAverage = \sum_{k=2}^{2N_g} k \cdot p_{x+y}(k)$$

- $p_{x+y}(k)$ adalah probabilitas jumlah indeks $i+j = k$

Sum Variance

- Definisi:
 - Mengukur penyebaran distribusi dari $p_{x+y}(k)$, yaitu sebaran penjumlahan indeks

$$SumVariance = \sum_{k=2}^{2N_g} (k - SumAverage)^2 \cdot p_{x+y}(k)$$

FEATURES ON CO-OCCURRENCE MATRIX

Sum Entropy

- Definisi:
 - Mirip dengan entropy, tetapi dihitung dari distribusi penjumlahan indeks $i+j$

$$\begin{aligned} & \text{SumEntropy} \\ &= \sum_{k=2}^{2N_g} p_{x+y}(k) \cdot \log_2(p_{x+y}(k) + \epsilon) \end{aligned}$$

Information Measures of Correlation 1 (IMC1)

- Definisi:
 - Mengukur ketergantungan informasi antara dua variable (I dan j dalam GLCM)
- $$IMC1 = \frac{HXY - HXY1}{\max(HX, HY)}$$
- HXY: entropy gabungan dari GLCM
 - HXY1: entropy gabungan berdasarkan marginal
 - HX, HY: entropy marginal (dari baris dan kolom)
- Interpretasi:
 - Nilai tinggi $\rightarrow$ korelasi informasi tinggi

FEATURES ON CO-OCCURRENCE MATRIX

Information Measures of Correlation 2 (IMC2)

- Definisi:
 - Alternatif dari IMC1, juga mengukur hubungan statistic antara dua piksel
- $$IMC2 = \sqrt{1 - \exp[-2(HXY2 - HXY)]}$$
- HXY2: bentuk alternatif dari entropy gabungan
- Interpretasi:
 - Nilai mendekati 1 → informasi sangat berkorelasi

THANK YOU